

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

(Level/CO) Marks

Q. 1 Solve Any Two of the following. 12

- A) A piece of equipment will function only when all the three components A, B, and C are working. The probability of failing of A during one year is 0.15, that of failing of B is 0.05 and that of C is 0.10. What is the probability that the equipment is will fail before the end of the year? L3/CO-1 6
- B) Find the probability that in a random arrangement of letters in the word "ASSASSINATION" the four S's come consecutively. L3/CO-1 6
- C) There are three bags; first contains 1 white, 2 red, 3 green balls; second contains 2 white, 3 red and 1 green and third contains 3 white, 1 red and 2 green balls. Two balls are drawn at random from a bag chosen at random. There are found to be 1 white and 1 red. Find the probability that balls came from Second bag. L3/CO-1 6

Q.2 Solve Any Two of the following. 12

- A) A random variable X has the following distribution: L4/CO-2 6

X:	1	2	3	4	5	6
P(X):	2k	4k	6k	8k	10k	12k

Determine (i) k (ii) $P(X < 4)$ (iii) $P(2 \leq X < 5)$ (iv) $E(X)$ (v) $\text{Var}(X)$

- B) A multiple choice test consists of 8 questions with 3 answers, with no negative marking for wrong answer. Student give answer randomly, to get distinction student must solve 75% correct answers. What is the chance that student must get distinction. L3/CO-2 6
- C) Fit a Poisson distribution to the following L2/CO-2 6

x	0	1	2	3	4
p(x)	46	38	22	9	1

Q. 3 Solve Any Two of the following.

12

- A) Calculate Karl Pearson's coefficient of correlation for the following data L2/CO-3 6

x	45	70	65	30	90	40	50	75	85	60
y	35	90	70	40	95	40	60	80	80	50

Calculate its probable error by assuming 60 and 65 as working means. Discuss whether or not calculated correlation coefficient is significant?

- B) If $r=0.8$, $\sum xy = 60$, $\sigma_y = 2.5$ and $\sum x^2 = 90$, where x and y are deviations from the respective means, compute the number of observations (n). L3/CO-3 6

- C) Calculate the rank correlation coefficient of the following data. L3/CO-3 6

x	68	64	75	50	64	80	75	40	55	64
y	62	58	68	45	81	60	68	48	50	70

Q.4 Solve Any Two of the following.

12

- A) Obtain regression line of y on x for the following data L2/CO-4 6

x	1	2	3	4	5	6	7	8	9
y	9	8	10	12	11	13	14	16	15

Also, obtain an estimate of y which should correspond on the average to $x=6.2$

- B) Given that for 100 items, the regression equations are $8x - 10y + 66 = 0$ and $40x - 18y = 214$. If variance of x is 9 then find L3/CO-4 6

- 1) Mean of x and y,
- 2) Correlation coefficient r_{xy} .
- 3) Standard deviation of y.

- C) For a sample of firms, the correlation between the profit rate (x) and growth rate (y) is found to be +0.9. Given the following additional information L3/CO-4 6

	Profit rate (%)	Growth rate (%)
Mean	20	25
Standard deviation	3	3

- 1) Compute the regression equation of y on x.
- 2) Estimate the likely growth rate (y) when profit rate is 16%.

Q. 5	Solve Any Two of the following.		12
A)	A coin was tossed 484 times and the head turned up 265 times. Test the hypothesis that the coin is unbiased at 5% LOS.	L2/CO-5	6
B)	In a city A, 20% of a random sample of 900 school boys had a certain slight physical defect. In another city B, 18.5 of a random sample of 1600 school boys had the same defect. is the difference between the proportion significance?	L3/CO-5	6
C)	A sample of 100 electric bulbs produced by manufacturer A showed a mean life time of 1190 hours and a standard deviation of 90 hours. A sample of 75 bulbs produced by manufacturer B showed a mean life time of 1230 hours with the standard deviation of 120 hours. Is there a difference between the mean life time of two hands at significance level of i)5% and ii)1%	L3/CO-5	6

*** Best of Luck ***